Basic SQL

BASIC SELECTS

```
SELECT first_name, last_name FROM Customers;
-- Selects first and last names of all customers

SELECT * FROM Vehicles WHERE size = 'SUV';
-- Selects all SUV-sized vehicles

SELECT email FROM Customers WHERE last_name = 'Smith';
-- Gets emails of customers with the last name Smith
```

FILTERING (`WHERE` , `BETWEEN` , `LIKE` , `IN` , `IS NULL`)

```
SELECT * FROM Rents WHERE rental_end_date BETWEEN '2024-01-01' AND
'2024-01-31';
-- Rentals ending in January 2024

SELECT * FROM Vehicles WHERE make ILIKE 'Ford%';
-- Vehicles where make starts with "Ford" (case-insensitive)

SELECT * FROM Pricing WHERE size IN ('SUV', 'Truck');
-- Prices for SUV or Truck sizes

SELECT * FROM Maintenance WHERE details IS NULL;
-- Maintenance records with no details
```

AGGREGATION (`COUNT` , `SUM` , `AVG` , `MAX` , `MIN`)

```
SELECT COUNT(*) FROM Customers;
-- Total number of customers

SELECT AVG(daily_rate) FROM Pricing;
-- Average daily rental rate across all sizes

SELECT MAX(rental_end_date) FROM Rents WHERE customer_id = 3;
-- Most recent rental end date for customer 3

SELECT SUM(amount) FROM Payments WHERE payment_method = 'Credit C
ard';
-- Total paid via credit card
```

GROUPING AND HAVING

```
SELECT size, COUNT(*) FROM Vehicles GROUP BY size;
-- Number of vehicles for each size

SELECT customer_id, COUNT(*)
FROM Rents
GROUP BY customer_id
HAVING COUNT(*) > 5;
-- Customers who made more than 5 rentals
```

JOINING TABLES

```
SELECT c.first_name, c.last_name, r.rental_start_date
FROM Rents r
JOIN Customers c ON r.customer_id = c.customer_id;
-- Rentals with corresponding customer names

SELECT v.make, v.model, l.location_name AS dropoff_location
```

```
FROM Rents r
JOIN Vehicles v ON r.vehicle_id = v.vehicle_id
JOIN Locations l ON r.dropoff = l.location_id;
-- Vehicle drop-off location details
```

ORDERING RESULTS

```
SELECT * FROM Pricing ORDER BY daily_rate DESC;
-- All prices ordered by daily rate (highest first)

SELECT * FROM Vehicles ORDER BY year ASC;
-- Oldest to newest vehicles
```

LIMITING RESULTS

```
SELECT * FROM Customers LIMIT 10;
-- First 10 customers

SELECT make, model FROM Vehicles ORDER BY year DESC LIMIT 1;
-- Newest vehicle in the fleet
```

SUBQUERIES

```
SELECT *
FROM Customers
WHERE customer_id IN (
    SELECT customer_id
    FROM Rents
    WHERE EXTRACT(YEAR FROM rental_start_date) = 2024
);
-- Customers who rented something in 2024
```

```
SELECT location_name
FROM Locations
WHERE location_id = (
    SELECT dropoff
    FROM Rents
    WHERE vehicle_id = 5
    ORDER BY rental_end_date DESC
    LIMIT 1
);
-- Last known dropoff location for vehicle 5
```

CASE STATEMENT

```
SELECT size,
    CASE
        WHEN daily_rate < 50 THEN 'Budget'
        WHEN daily_rate < 100 THEN 'Standard'
        ELSE 'Premium'
    END AS price_category
FROM Pricing;
-- Categorizes pricing tiers based on daily rate
```

DISTINCT VALUES

```
SELECT DISTINCT size FROM Vehicles;
-- All unique vehicle sizes available

SELECT DISTINCT payment_method FROM Payments;
-- All types of payment methods used
```

DATE FUNCTIONS

```
SELECT EXTRACT(MONTH FROM rental_start_date) AS month, COUNT(*)
FROM Rents
GROUP BY month;
-- Rentals per month (across all years)

SELECT * FROM Payments
WHERE payment_date >= CURRENT_DATE - INTERVAL '30 days';
-- Payments made in the last 30 days
```

OTHER DOMAINS

School / University

```
SELECT name FROM Students WHERE gpa > 3.5;
-- Students with GPA above 3.5

SELECT course_id, COUNT(*) FROM Enrollments GROUP BY course_id;
-- Number of students per course
```

E-Commerce

```
SELECT name, price FROM Products WHERE price < 100;
-- Products cheaper than 100

SELECT customer_id, SUM(total) FROM Orders GROUP BY customer_id;
-- Total spending per customer
```

HR System

```
SELECT department, AVG(salary) FROM Employees GROUP BY department;
-- Average salary per department

SELECT * FROM Employees WHERE hire_date BETWEEN '2020-01-01' AND '2
```

```
022-12-31';
-- Employees hired between 2020 and 2022
```

1. List all rentals with customer name and vehicle info

```
SELECT
   c.first_name,
   c.last_name,
   v.make,
   v.model,
   r.rental_start_date,
   r.rental_end_date
FROM
   Rents r
JOIN
   Customers c ON r.customer_id = c.customer_id
JOIN
   Vehicles v ON r.vehicle_id = v.vehicle_id;
-- Shows who rented which car and when.
```

---

2. Find each customer's most recent dropoff location

```
SELECT
   c.first_name,
   c.last_name,
   l.location_name AS last_dropoff
FROM
   Customers c
JOIN
   Rents r ON c.customer_id = r.customer_id
JOIN
   Locations l ON r.dropoff = l.location_id
WHERE
   r.rental_end_date = (
```

```
        SELECT MAX(r2.rental_end_date)
        FROM Rents r2
        WHERE r2.customer_id = c.customer_id
    );
-- Finds the latest dropoff location for each customer.
```

---

3. List vehicles that have never been rented

```
SELECT *
FROM Vehicles v
WHERE NOT EXISTS (
    SELECT 1
    FROM Rents r
    WHERE r.vehicle_id = v.vehicle_id
);
-- Filters vehicles that have no entry in Rents table.
```

---

4. Show all dropoffs for vehicles currently under maintenance

```
SELECT DISTINCT l.location_name
FROM Maintenance m
JOIN Rents r ON m.vehicle_id = r.vehicle_id
JOIN Locations l ON r.dropoff = l.location_id
WHERE m.status = 'Scheduled';
-- Where the vehicle under maintenance has previously been dropped off.
```

---

5. Find customers who have rented from multiple locations

```
SELECT customer_id
FROM Rents
GROUP BY customer_id
```

```
HAVING COUNT(DISTINCT pickup) > 1;
-- Customers who picked up cars from more than one location.
```

🔷 Subquery Examples (With explanation)

6. Get the most expensive daily rate

```
SELECT *
FROM Pricing
WHERE daily_rate = (
    SELECT MAX(daily_rate) FROM Pricing
);
-- Finds the vehicle size with the highest daily rate.
```

7. List customers who have rented a ‘Luxury’ car

```
SELECT DISTINCT c.first_name, c.last_name
FROM Customers c
WHERE c.customer_id IN (
    SELECT r.customer_id
    FROM Rents r
    JOIN Vehicles v ON r.vehicle_id = v.vehicle_id
    WHERE v.size = 'Luxury'
);
-- Only customers who’ve rented a Luxury vehicle at least once.
```

8. Show vehicles rented in the same month they were added to maintenance

```
SELECT v.vehicle_id, v.make, v.model
FROM Vehicles v
WHERE EXISTS (
    SELECT 1
```

```
    FROM Rents r
    JOIN Maintenance m ON m.vehicle_id = r.vehicle_id
    WHERE r.vehicle_id = v.vehicle_id
    AND EXTRACT(MONTH FROM r.rental_end_date) = EXTRACT(MONTH FRO
M m.maintenance_date)
);
-- Vehicles where rental and maintenance happened in same month.
```

9. List customers with total payments above average

```
SELECT c.first_name, c.last_name
FROM Customers c
JOIN (
    SELECT customer_id, SUM(amount) AS total_paid
    FROM Payments
    GROUP BY customer_id
) p ON c.customer_id = p.customer_id
WHERE p.total_paid > (
    SELECT AVG(amount) FROM Payments
);
-- Only customers who paid more than the average payment amount.
```

10. Most used pickup location for ‘Fullsize’ cars

```
SELECT l.location_name, COUNT(*) AS pickup_count
FROM Rents r
JOIN Vehicles v ON r.vehicle_id = v.vehicle_id
JOIN Locations l ON r.pickup = l.location_id
WHERE v.size = 'Fullsize'
GROUP BY l.location_name
ORDER BY pickup_count DESC
LIMIT 1;
-- Most frequent pickup spot for Fullsize cars.
```

Bonus: Nested JOIN + Aggregation + Subquery

11. Top 3 customers by total payments

```
SELECT c.first_name, c.last_name, SUM(p.amount) AS total_paid
FROM Payments p
JOIN Customers c ON p.customer_id = c.customer_id
GROUP BY c.customer_id, c.first_name, c.last_name
ORDER BY total_paid DESC
LIMIT 3;
-- Shows the 3 highest-paying customers.
```